(Exam) (i) 512 x 512 sub image out from the center of an 800x600 image. what are the coordinate of the pixel that in at the lower left corner...

Ans: Given, large image = 800x600.

center = $\left(\frac{800}{2}, \frac{600}{2}\right) = (400, 300)$

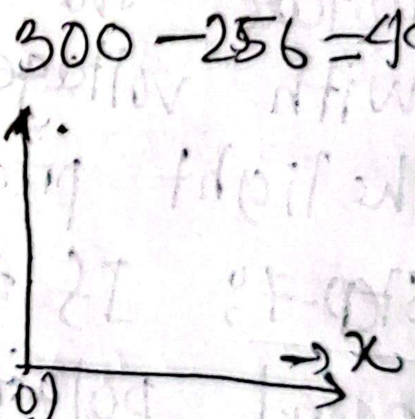
small image = (512x512)

center = $\frac{512}{2}$ = (256x256)

lower left corner $\Rightarrow x = 400-256 = 144$.

$y = 300-256 = 44$.

so, (144, 44)



upper left corner,

$x = 400-256 = 144$ (0,0)

$y = 300+256 = 556$.

- Lower right; $x = 400+256 = 656$

$y = 300-256 = 44$

→ ~~tef~~

upper ~~left~~ right = $x = 400+256 =$

$y = 300+256 =$

(ii) → Find CMY ... RGB (0.2, 1, 0.5).

$C = 1 - R$, $M = 1 - G$, $Y = 1 - B$.

$C = 1 - 0.2$ $\quad M = 1 - 1$, $\quad Y = 1 - 0.5$.

$= 0.8$ $\quad = 0$ $\quad = 0.5$.

CMY = (0.8, 0, 0.5).

(iii) RGB direct coding with 2-bits per ...

total bit per pixel = 2+2+2 = 6-bit.

no. of possible color = $2^6 = 64$.

(iv) What is the color of CRT.

pitch

→ The pitch color of CRT usually means a dot pitch. It is the distance between two adjacent phosphor dots or between " " pixel triads on the CRT. screen

(v) Why do many color printer use black pigment — color printer use CMY color — cyan, magenta, & yellow. Ideally mixing C, M, & Y should produce Black. But in practice, the mixture doesn't produce pure black. It often produce muddy dark brown or gray color.

Reason — 1. Produces pure & sharp black
2. Make text clearer.
3. Save color ink.
4. Reduce printing cost.

Exercise (ch-02)

(2.3) Resolution of a 2x2 inch image that has 512 x 512 pixels.

→ $\frac{512}{2} = 256$ pixels per inch.

(2.4) Image aspect ratio :- The ratio of it's width to it's height, measured in unit length or number of pixels.

(2.5) Given, Height = 2 inch. Aspect ration = 1.5, width = ?

$\therefore$ width = 1.5 × Height = 1.5 × 2 = 3 inche

(2.6) To resize 1024 × 768 image to one that is 640 pixels wide with the same aspect ration, what would be the height?

→ Same aspect ratio means the image shape must not be distorted.

- shape যেন change না হয়. — original width = 1024, New " = 640

- scale factor — $\frac{640}{1024} = 0.625$.

যেমন height ও same ratio তে কমবে.

$$\text{Scale factor} = \frac{\text{new height}}{\text{original}}$$

$$\Rightarrow 0.625 = \frac{\text{new}}{768}$$

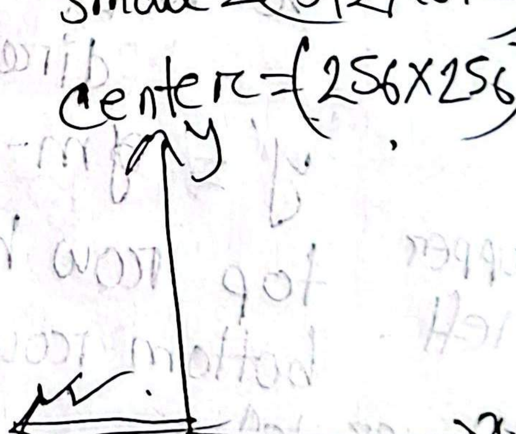
$$\Rightarrow \text{new} = 768 \times 0.625 = 480$$

∴ The resized image is $= 640 \times 480$.

(2.7) To cut 512×512 sub image out from the ~~corner~~ center of an 800×600 image, coordinate at lower left corner.

→ large image $= (800 \times 600)$, center $= (400 \times 300)$

small $= (512 \times 512)$, center $= (256 \times 256)$



→ ~~40~~ $x = 400 - 256 = 144$.

$y = 300 - 256 = 44$

∴ [144, 44]

(2.8) Coordinate conversion; upper-left origin to ~~to~~ lower-left origin:

— upper-lef origin system:→

$(0,0) \dashrightarrow x$

$\downarrow$

y

→ mathematical

$y \uparrow$

$0,0 \longrightarrow x$

as we go upward y increase.

~~Hav~~

— Given a pixel coordinate in upper-left origin system: (x, y).

convert it to lower-left ~~corner~~ origin system.

$$\left[(x', y') = (x, m - y - 1) \right]$$ m = no. of pixel in vertical direction.

$x' = x$ [The horizontal direction doesn't change]

$y' = m - y - 1$ (vertical direction reversed)

upper left. top row has $y = 0$.
bottom row has, $y = m-1$.

lower-left → " " " $y' = 0$
top " " $y' = m-1$. (-1 কারন pixel counting 0 থেকে)

(2.9) Find CMY coordinate -- RGB space.
(0.2, 1, 0.5)

- $CMY = 1 - (RGB)$

$$= (1-0.2,\ 1-1,\ 1-0.5)$$

$$CMY = (0.8,\ 0,\ 0.5)$$

(2.10) RGB = ? if CMY space = (0.15, 0.75, 0)

$$RGB = (1-0.15,\ 1-0.75,\ 1-0)$$

$$= (0.85,\ 0.25,\ 1)$$

(2.11) If we use direct coding of RGB with 2-bit per primary color-values

- possible colors we have for each pixel

$$= 2^2 \times 2^2 \times 2^2 = 4 \times 4 \times 4 = 64.$$

(2.12) with 10-bit primary color of RGB

$$= 2^{10} \times 2^{10} \times 2^{10} = 1024^3$$

(2.13) RGB direct coding.

Red = 5-bit for each, Blue = 6-bits

Total ~~Green~~ = 16-bits per pixels.

— no. of possible = $2^5 \times 2^{5} \times 2^6$

$= 2^{16}$

5, 6 = 16,
⇒ = 5.

(2.14) If we use 12-bit pixel value - how many entry does the lookup table have.

→ If pixel value = 12-bit.

than possible index value = 2^{12}

= 4096. entries.

(2.15) 2-byte pixel value &
24-bit lookup table representation
size of the lookup table = ?

So, 1 byte = 8-bit ∴ 2-byte = 16-bit.

means pixel value is 16-bit.

no. of entries = 2^{16} = 65536.

Each entry stores 24-bit color

24-bit = $\frac{24}{8}$ = 3-bytes.

so each lookup table entry needs = 3 bytes

Total size = 65536 × 3 = bytes.

(2.16) Fake statement →

Flurocence means → light is emitted while the phosphor is being exposed to the electron beam. So electron beam is hitting the phosphors, & the same time light is produced.

Phosphorescence means light ~~is~~ continues to be emitted ~~by~~ after the electron beam exposure has stopped.

(2.17) Persistence → The duration of phosphorescence exhibited by a phosphor.

(2.18) Control electrode in a CRT → function is – Regulate the intensity of the electrode beam.

(2.19) Two method by which an electron beam can be bent?

1) Electrostatic deflection.
2) magnetic deflection.

(2.20) Horizontal retrace :→ The path the electron beam takes when returning to the left side of the CRT screen.

(2.21) Vertical retrace - The path the electron beam takes at the end of each refresh cycle.

(2.22) The pitch of a color CRT → The distance between the center of the phosphor dot patterns on the inside of the display screen.

(2.23) color printer normally use three color pigment →. C = cyan. M = Magenta Y = Yellow.

Theoretically - If cyan, magenta & yellow are mixed together, they

should produce Black. :- $C+M+Y = \text{Black}$.
But practically, this mixture does not produce high quality pure black. It often produce a muddy dark brown..
So printer use a seperate black pigment called-K — $CMY + K = CMYK$.
- Black pigment gives pure black. Text become sharper & clearer.

(2.24) ~~Given~~ suppos we have - $n \times n$-pixel grid
So ~~total~~ no. of pixel $= n \times n = n^2$.
- Each pixel can take m intensity levels.
- so there are m level $0, 1, 2, 3 \ldots, m-1$ totals. But among them - 0 means zero intensity or pixel OFF.
So the no. of non zero intensity levels for $(m-1)$.
- ~~total~~ overall intensity now $= n^2(m-1)$
কিন্তু সব pixel off থাকলে একটা zero intensity level পাওয়া যায়। তাই আরও 1 যোগ করতে হয়

So, $\Rightarrow n^2(m-1)+1$.

(2.25) Dither matrix tells the order in which pixels are turned ON. to represent different intensity level.

$$\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix}$$ — A 2×2 matrix has 4 position (0, 1, 2, 3)

— The smaller values turned ON first.

(2.26)